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Investigating the Bilingual Aging Lexicon: A Network Analysis of Word Associations in Chinese–English Bilinguals

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ABSTRACT

The present study examines the bilingual aging lexicon by comparing word association networks across four age groups of Chinese–English bilingual adults. Adopting network analysis with breadth-first search-based sampling, results showed that in L1 (Chinese) networks, global connectivity increases from early to mid-adulthood, accompanied by reduced local clustering and enhanced global efficiency; the network remains relatively stable through midlife, followed by declining efficiency and progressively pronounced modular structure in late adulthood. The L2 (English) networks mirror L1 developmental trends in early and mid-adulthood but with substantially larger shifts; throughout adulthood, L2 exhibits lower global and local connectivity and weaker modularity compared to L1. These findings reveal distinct developmental trajectories for L1 and L2 semantic networks in bilinguals, underscoring greater structural stability in the dominant language (L1) across adulthood relative to the nondominant language (L2). This L1–L2 asymmetry points to differential impacts of aging on bilingual lexical organization. Results are discussed in relation to compensatory mechanisms and enrichment accounts of cognitive aging.

Keywords: Aging lexicon; Bilingual adult; Semantic network; Word association

1. Introduction

Network science provides a framework for modeling cognitive and linguistic systems, including the mental lexicon, as semantic networks where nodes represent words or concepts and edges indicate their associations (De Deyne, Kenett, Anaki, Faust, & Navarro, 2017). These networks are often derived from word association tasks, in which participants produce

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responses to cue words (De Deyne & Storms, 2008; Dubossarsky, De Deyne, & Hills, 2017; Li, Jiang, Shang, & Chen, 2021; Martinez & Vitevitch, 2024). Network metrics reveal structural features of the mental lexicon that relate to cognitive processes. For example, average degree ($<k>$) reflects the average number of connections per node; lower average degree indicates sparser networks with fewer connections overall, which can lead to less efficient information diffusion and retrieval, as activation spreads more slowly across a less interconnected structure (Wulff, Hills, & Mata, 2022). Average shortest path length (L) measures the average number of steps between nodes; longer paths indicate lower global efficiency, potentially slowing semantic retrieval and access to related concepts, as information must traverse more intermediary nodes (Cosgrove, Beaty, Diaz, & Kenett, 2023; Kenett, Anaki, & Faust, 2014). Clustering coefficient (C) measures the extent to which neighbors of a word are also neighbors of each other. In phonological networks, high clustering slows recognition by trapping activation among neighbors (Chan & Vitevitch, 2009), yet facilitates long-term word learning by reinforcing local structure (Goldstein & Vitevitch, 2014); in semantic networks, higher clustering strengthens local connectivity, facilitating the efficient spread of activation within tightly knit clusters of related concepts. This local structure supports faster processing for closely related items, as activation is more likely to remain concentrated and reinforce associations among neighboring nodes, leading to greater priming effects for semantically associated words (Siew, 2019), although sometimes this may come at the cost of reduced flexibility in accessing distant concepts, as activation can be confined in dense local modules. Modularity (Q) quantifies the division of networks into communities; higher modularity implies more segregated modules that may enhance specialized knowledge but reduce overall cognitive flexibility, as the network becomes more rigid and compartmentalized, limiting the ability to form novel connections across modules during tasks like creative problem-solving or divergent thinking (Campidelli, Domanti, Fusi, Kenett, & Agnoli, 2026; Kenett et al., 2014, 2016).

Research on age-related changes in semantic networks often reports that networks derived from older adults' responses appear structurally sparser than those of younger adults, with lower average degree, higher average path length, lower clustering coefficients, and higher modularity (Cosgrove, Kenett, Beaty, & Diaz, 2021; Dubossarsky et al., 2017; Wulff et al., 2022). These patterns have traditionally been interpreted as reflecting age-related decline in lexical access and reduced efficiency in spreading activation, consistent with broader reductions in fluid cognitive abilities. More recent work, however, proposes an alternative interpretation in which these patterns arise not from degradation but from cognitive network enrichment resulting from lifelong learning (Hills, 2025a, 2025b). Under this account, continued experience expands the mental lexicon by adding nodes and connections, increasing vocabulary and crystallized knowledge. Paradoxically, this enrichment can produce behavioral patterns that resemble sparsening in derived networks. As the number of potential associations increases, competition among candidates during retrieval also increases, leading to higher response entropy and less predictable associations. Consequently, free association data may show reduced overlap across participants, yielding networks that appear less densely connected even when the underlying representation is more richly interconnected. This distinction highlights an important methodological issue: semantic networks derived from behavioral tasks may reflect both the structure of lexical representations and the

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processes involved in retrieving associations. In the present study, we focus on the structural properties of word association networks across age groups in bilingual speakers. While our analyses cannot fully disentangle representational enrichment from age-related changes in retrieval dynamics, examining patterns across languages and age groups can provide insight into how lifelong bilingual experience shapes the organization of the mental lexicon.

Beyond age-related changes in monolingual networks, recent research has increasingly focused on cognitive development in bilingual adults and how bilingual experience interacts with aging. However, our understanding of the bilingual aging lexicon remains limited, particularly regarding structural differences between first (L1) and second (L2) language networks within the same individuals. One relevant line of research comes from Borodkin, Kenett, Faust, and Mashal (2016), who used semantic fluency data and small-world network analysis to compare L1 and L2 networks in English–Hebrew bilinguals. In a semantic fluency task, participants generate words belonging to a specified category (e.g., animals), which differs methodologically from free word association tasks, where responses are not constrained by category membership. Because semantic fluency responses may partly reflect category-based search strategies in addition to underlying lexical structure, findings derived from this paradigm cannot be assumed to generalize directly to networks constructed from word association data. Nevertheless, such studies provide useful insights into how bilingual experience may shape lexical organization. Borodkin et al. (2016) reported that L2 networks exhibited significantly higher clustering coefficients and lower modularity than both the bilinguals' own L1 networks and those of native Hebrew speakers. These differences persisted in a subsample where L2 vocabulary size was matched to that of native speakers, suggesting that structural variation may arise from factors such as later age of acquisition and reduced immersion rather than lexicon size alone. Differences in average shortest path length were inconsistent across analyses. Other reports, such as Agustín-Llach and Palapanidi (2024), suggest that L2 networks may be sparser, with higher average path lengths, lower clustering, and higher modularity—potentially reflecting less exposure or L1 interference. However, these results are based primarily on descriptive statistics without inferential testing, limiting their value for hypothesis development. Taken together, these studies suggest that language proficiency and language experience may shape the structural organization of bilingual lexical networks. Such differences may have broader implications for bilingual cognition, including well-documented patterns such as enhanced executive control alongside potential costs in lexical access speed (Bialystok, 2016).

Despite these insights, little is known about age-related changes in bilingual semantic networks in mid- to late adulthood, particularly among high-proficiency bilinguals with advanced education. This group offers a useful case for examining how lifelong bilingual experience interacts with aging to influence network structure and related cognitive processes. The present study addresses this by analyzing semantic networks from a word association task with Chinese university English teachers aged 30–70. We ask:

1. How do network metrics (average degree, average path lengths, clustering coefficient, modularity) change with age in bilingual adults' L1 (Chinese) and L2 (English) word association networks?

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2. Do age-related changes in network metrics differ between L1 and L2?

Based on prior findings and the network enrichment account (Dubossarsky et al., 2017; Hills, 2025, 2025ab; Wulff et al., 2022), we hypothesize that older adults' networks will show lower average degrees, higher average path lengths, lower clustering coefficients, and higher modularity in both languages, reflecting the combined influences of accumulation and reorganization of vocabulary knowledge and cognitive decline. We also tentatively predict that L2 lexical networks may exhibit higher clustering coefficients and lower modularity than L1 networks, in line with patterns reported in Borodkin et al. (2016), while acknowledging that those findings were derived from semantic fluency data rather than word association tasks. No directional hypotheses are made for average degree or average shortest path length due to insufficient consistent evidence.

2. Methods

2.1. Participants

A total of 117 in-service English teachers (female = 78, male = 39) were recruited from universities across Jiangsu Province, China. Educational attainment included bachelor's ($n = 25$, 21.4%), master's ($n = 16$, 13.7%), and doctoral degrees ($n = 75$, 64.1%), with 29 participants (24.8%) reporting experience of studying abroad. Participants were stratified into four age groups following standard decadal classifications in aging research: 30–40 years ($n = 30$, $M = 36.23$, $SD = 2.81$), 41–50 years ($n = 30$, $M = 42.73$, $SD = 2.64$), 51–60 years ($n = 30$, $M = 54.53$, $SD = 2.91$), and 61–70 years ($n = 27$, $M = 65.04$, $SD = 2.43$). No participants reported neurological disorders or cognitive impairments.

2.2. Materials

The study employed a continued word association task using 30 English and 30 Chinese stimulus words originally selected by Zhang (2010). The 30 English cue words were drawn from the 1000 and 2000 word lists in Nation's (1990) frequency lists to ensure high familiarity among learners. A recheck with Nation's (2016) updated BNC/COCA lists showed that 27 of these words still fall within the top 2000 frequency bands, while three words (imply, essential, and complex) now appear in the 3000-word list. This consistency suggests that the original selection remains highly representative of high-frequency and widely used vocabulary even after more than a decade. To mitigate potential confounding variables, the stimuli were matched on part-of-speech category (nouns, verbs, adjectives), concreteness (concrete vs. abstract), and word length/strokes. For English words, a paired-samples t-test revealed a significant difference in concreteness ratings between concrete and abstract items, $t(29) = 39.55$, $p < .001$. In contrast, ANOVA indicated no significant difference in concreteness across the three parts of speech, $F(2, 27) = 0.03$, $p = .971$. Nonparametric tests further confirmed the absence of significant differences in word length between concrete and abstract stimuli (Mann–Whitney $U = 97.50$, $z = -0.63$, $p = .526$) and across parts of speech (Kruskal–Wallis $H(2) = 1.06$, $p = .590$). For the Chinese stimuli,

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a Kruskal–Wallis test showed no significant difference in strokes across the three parts of speech, $H(2) = 5.28$, $p = .071$. Similarly, concreteness ratings did not differ significantly across the three categories, $H(2) = 2.39$, $p = .302$. When comparing concrete and abstract Chinese items, Wilcoxon rank-sum tests revealed a significant difference in concreteness, $W = 121.00$, $z = -4.63$, $p < .001$, with a large effect size ($r = -.844$). In contrast, no significant difference emerged in strokes between concrete and abstract items ($W = 219.50$, $z = -0.54$, $p = .590$). See Appendix S1 for the complete stimulus word list.

2.3. Procedure

Data for the present study were collected in the second half of 2024. Since the participants were all university teachers, it was not possible to collect data in a group setting like with students. We, therefore, built an online applet for data collection. Before the main study, several pilot tests were implemented to improve the order of words, the instructions, font size, and line spacing, which made the task more reliable and easier to use. For accessibility purposes, the applet also provided an optional voice-input function for participants who found typing inconvenient. In such cases, spoken responses were automatically converted into text before submission. The system stored only the final textual responses and did not record the specific input modality used for each response.

Each stimulus word was shown on its own page. The instructions for the English task were: "In the following test, an English word will appear on each of the 30 pages. When you see the word, please write down the first three English words that come to your mind right away. Do not think too long. The label in parentheses indicates their part of speech: *n.* = noun; *v.* = verb; *adj.* = adjective." The Chinese instructions were the same, but translated carefully to keep the meaning consistent. There was no time limit, so participants could answer at their own pace and give spontaneous responses, and they needed to enter exactly three words for each stimulus before moving to the next page. The participants completed the English word association task first and did the Chinese task about 6 months later. Informed consent was obtained from all participants before they started.

2.4. Analysis

As participants were instructed to provide three valid responses for each stimulus, and the online interface enforced this rule by preventing progression until all three fields were completed, no responses were blank, skipped, or excluded during data collection. As a result, the 30–40, 41–50, and 51–60 age groups each contributed 2700 valid associations per language (30 participants × 30 stimuli × 3 responses). The 61–70 age group included 27 participants due to recruitment constraints with this age range, yielding 2430 responses per language. All data were retained to maintain the authenticity of the response distribution.

Prior to analysis, responses were carefully normalized to ensure cross-group comparability. For the English data, all responses were standardized to American English spelling (e.g., *color* instead of *colour*) to eliminate orthographic inconsistencies. Minor typographical or spelling errors resulting from accidental input mistakes were corrected (e.g., *moutain* → *mountain*). Inflected forms (e.g., *loves*, *loved*) were lemmatized to their base forms (e.g., *love*) so that

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different morphological variants would not be treated as separate nodes in the semantic network. For Chinese responses, adjectival responses were normalized by removing the structural particle "的" (*de*), which carries no semantic weight in this context (e.g., 漂亮的 [*piàoliang de*, beautiful] → 漂亮 [*piàoliang*, beautiful]). This step ensured equivalence between forms with and without the particle.

Following established methodologies (Coronges, Stacy, & Valente, 2007; Zortea, Menegola, Villavicencio, & de Salles, 2014), we constructed response–response (R–R) networks to model the bilingual aging lexicon, prioritizing this approach for its ability to capture productive lexical organization from freely generated words. In R–R networks, nodes represent unique response words, and edges reflect co-occurrence when two responses appear under the same cue. Edge weight equals the number of distinct cues in which a pair co-occurred. Within each cue, responses were treated at the type level, meaning that duplicate responses were removed so that each response word appeared only once per stimulus. Consequently, a pair of responses could contribute at most one co-occurrence per cue, and higher edge weights indicate that the two concepts were jointly produced under multiple different cues, reflecting stronger contextual similarity across stimuli. For example, in the 30–40 age group, *air* and *rain* co-occurred once (under *DRY*; weight = 1), *body* and *heart* twice (*MENTAL*, *SOFT*), and *cold* and *water* three times (*DRY*, *RIVER*, *HOT*), with higher weights signaling stronger contextual similarity. We selected R–R over stimulus–response (S–R) or stimulus–stimulus (S–S) networks to ensure ecological validity and analytical power. S–R networks, while capturing direct associative strength, require repeated testing rounds—impractical with older adults. S–S networks, limited to our 30 cue words, produce small graphs ($\leq$ 30 nodes) that primarily reflect associative domain overlap among cues (i.e., shared response profiles). In contrast, R–R networks can effectively represent the productive lexicon, and this type of co-occurrence network construction method aligns with broader network science traditions in semantic, conceptual, and informational networks (Marjieh et al., 2025; Siew, Wulff, Beckage, & Kenett, 2019).

We computed key graph-theoretic metrics to characterize word association network structure, building on prior work linking these indices to cognitive processes (Cosgrove et al., 2023; Kenett et al., 2014; Siew, 2019; Wulff et al., 2022). In summary, average degree ($<k>$) reflects global connectivity; lower values indicate sparser networks with slower information diffusion (Wulff et al., 2022). Average shortest path length (L) measures global efficiency; longer paths imply reduced retrieval speed, as activation traverses more steps to reach distant concepts (Cosgrove et al., 2023). Clustering coefficient (C) quantifies local connectivity by measuring the extent to which neighbors of a node are also connected to each other. Higher clustering reflects densely interconnected local neighborhoods, which can facilitate rapid activation spread among closely related concepts and support efficient processing of semantically associated items (Siew, 2019). Modularity (Q) assesses community-level segregation by quantifying the extent to which the network is divided into relatively distinct modules. Higher modularity indicates stronger separation between communities, reflecting a more compartmentalized lexical organization that may constrain flexible access across semantic domains (Kenett et al., 2014, 2016).

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Network metrics, including n, m, $<k>$, L, C, and Q, were computed using *NetworkX*, a Python package for complex network analysis (Hagberg, Schult, & Swart, 2008). Modularity was calculated via the Louvain algorithm (Blondel, Guillaume, Lambiotte, & Lefebvre, 2008) implemented in *NetworkX*'s community module. To maintain the ecological validity of the semantic networks, no artificial control was applied to the number of nodes when constructing and reporting the original networks for each age/language group. However, for the statistical comparison involving repeated subgraph sampling, strict control of node counts was implemented to guarantee comparability and statistical rigor. During this phase, we adopted a breadth-first search (BFS)–based subgraph sampling approach rather than node-resampling or participant-level bootstrapping. Node-resampling methods (e.g., Efron-style bootstrapping) are often applied in semantic fluency studies, where networks are dense and category-based, but they may fragment sparser word association networks and distort connectivity. Participant-level bootstrapping (Christensen & Kenett, 2023) is more appropriate when individual-level networks can be reliably estimated, which is not feasible for single-round word association data. In contrast, BFS sampling preserves local connectivity and clustering structure while providing size-controlled subnetworks for structural comparisons across groups (Leskovec & Faloutsos, 2006; Maiya & Berger-Wolf, 2011).

Starting from randomly selected seed nodes, BFS iteratively expanded through neighboring nodes until reaching the target subgraph size ($n = 1000$), with 100 replicates per network. In prior network-sampling research, BFS subgraphs typically include about 10–30% of the original nodes, a range shown to preserve key topological properties (Leskovec & Faloutsos, 2006; Maiya & Berger-Wolf, 2011). However, these guidelines are primarily based on very large networks with thousands or millions of nodes. In comparatively sparse word association networks, smaller samples can easily fragment the graph and distort connectivity patterns, particularly when co-occurrence relationships depend on a limited set of cue words. Larger subgraphs, therefore, provide greater structural representativeness and help preserve the global connectivity and modular organization of the original network. Accordingly, the present study used a relatively large subgraph size—approximately 70% of the original network—to balance structural fidelity with statistical replication. This choice ensured that the extracted subnetworks retained the major connectivity patterns of the full networks while still allowing repeated sampling for statistical comparison. Importantly, the goal of this BFS-based sampling procedure was not to approximate different cue-word samples but to generate size-controlled subnetworks that preserved the underlying response–response co-occurrence structure. Because edges in R–R networks depend on responses co-occurring across multiple cues, reducing the number of cues during network construction would substantially limit co-occurrence opportunities and often lead to fragmented graphs with unstable connectivity patterns. By contrast, BFS sampling maintains the full cue structure while enabling controlled comparisons across age groups and languages. Although R–R networks often exhibit limited edge overlap across groups due to age-related associative differences, the resulting structural metrics (e.g., degree, clustering) remain comparable because they quantify relative topology independent of specific lexical content—provided that cues, instructions, and network scale are consistent (as in Zortea et al., 2014; Jeong & Hills, 2024; Martinez & Vitevitch, 2024). To further increase transparency, we also quantified the degree of response overlap using

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Table 1
Network metrics across age groups and languages

Network metrics	30–40		41–50		51–60		61–70	
	Cn	En	Cn	En	Cn	En	Cn	En
n	1313	1044	1350	1110	1349	1196	1418	1212
m	38,913	27,068	42,944	35,081	42,135	35,153	43,919	37,784
<*k*>	59.27	51.85	63.62	63.21	62.47	58.78	61.94	62.35
L	2.565	2.406	2.421	2.115	2.382	2.326	2.495	2.264
C	0.939	0.916	0.930	0.897	0.935	0.917	0.945	0.912
Q	0.746	0.686	0.721	0.598	0.692	0.718	0.767	0.653

Abbreviations: *C*, clustering coefficient; <*k*>, average degree; *L*, average shortest path length; *m*, number of edges; *n*, number of nodes; *Q*, modularity.

Table 2
Descriptive statistics (*M* and *SD*) for subgraph metrics across age groups and languages

Metric	30–40		41–50		51–60		61–70	
	Cn	En	Cn	En	Cn	En	Cn	En
<*k*>	58.943	51.591	63.105	62.827	62.293	57.746	61.152	61.481
	(0.767)	(0.359)	(0.883)	(0.599)	(0.962)	(0.579)	(0.931)	(0.631)
L	2.410	2.388	2.312	2.081	2.259	2.281	2.362	2.208
	(0.033)	(0.006)	(0.025)	(0.014)	(0.028)	(0.018)	(0.036)	(0.019)
C	0.937	0.914	0.927	0.892	0.932	0.913	0.942	0.906
	(0.010)	(0.003)	(0.007)	(0.006)	(0.008)	(0.010)	(0.010)	(0.008)

both overall vocabulary and cue-level measures, with detailed results provided in a separate OSF document (Word Association Response Overlap Across Age Groups). This Supplementary Analysis documents the age-related differences in response words without affecting the main structural comparisons. Structural comparisons were conducted on size-controlled subnetworks to ensure metric reliability independent of differences in vocabulary size. For each sampled subgraph, we computed average degree (<*k*>), average shortest path length (*L*), clustering coefficient (*C*), and modularity (*Q*). The resulting metric distributions across the 100 replicates were then subjected to two-way ANOVAs or nonparametric tests to examine differences across age groups and languages.

All datasets and analysis scripts used in this study are available on OSF at https://osf.io/t6wsb/.

3. Results

Descriptive results for original networks and subnetworks are summarized in Tables 1 and 2, respectively. Note that the original network metrics are reported for reference but vary due to unequal node and edge counts. Key results and interpretations are based on

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statistical comparisons using 1000-node BFS subgraphs, which control for network size and enable valid group differences.

For average degree (<k>), Chinese networks have consistently higher <k> than their English counterparts across all age groups. The two-way ANOVA revealed significant main effects of age, $F(3, 792) = 4000.397$, $p < .001$, $\eta^2_p = 0.938$, language, $F(1, 792) = 3202.462$, $p < .001$, $\eta^2_p = 0.802$, and a significant interaction, $F(3, 792) = 1210.992$, $p < .001$, $\eta^2_p = 0.821$. Post hoc pairwise comparisons with Bonferroni correction further clarified these effects. In the Chinese (L1) networks, <k> increased from 30–40 to 41–50, reaching a peak ($t(198) = -35.57$, $p < .001$, Cohen's $d = 5.03$). Among the older three groups, <k> declined gradually from 41–50 to 51–60 ($t(198) = 6.21$, $p < .001$, $d = 0.88$), and further to 61–70 ($t(198) = 8.52$, $p < .001$, $d = 1.21$). Notably, the largest effect size among all L1 comparisons was between 30–40 and 41–50 ($d = 5.03$). The English (L2) networks exhibited a more dynamic trajectory, but peaking again in the 41–50 group ($t(198) = -160.90$, $p < .001$, $d = 22.75$). It then declined to 51–60 ($t(198) = 60.96$, $p < .001$, $d = 8.62$) and rose again in 61–70 ($t(198) = -43.58$, $p < .001$, $d = 6.16$). Despite this rebound, the 61–70 group remained below the 41–50 group ($t(198) = 15.46$, $p < .001$, $d = 2.19$). Importantly, all effect sizes among the three older groups were smaller ($d = 2.19$–8.62) than those comparing 30–40 with any older group ($d = 12.76$–22.75). Regarding between-language comparisons, Chinese networks exhibited higher <k> than English networks in four groups, although the effect sizes in 41–50 and 61–70 group were small: 30–40 ($t(198) = 86.80$, $p < .001$, $d = 12.28$), 41–50 ($t(198) = 2.61$, $p = .039$, $d = 0.37$), 51–60 ($t(198) = 40.48$, $p < .001$, $d = 5.72$), and 61–70 ($t(198) = -2.92$, $p = .016$, $d = 0.41$).

For average shortest path length (L), English networks have consistently lower L than their Chinese counterparts, except in the 51–60 age group. The two-way ANOVA revealed significant main effects of age, $F(3, 792) = 2356.417$, $p < .001$, $\eta^2_p = 0.899$, language, $F(1, 792) = 3093.775$, $p < .001$, $\eta^2_p = 0.796$, and a significant interaction, $F(3, 792) = 1157.956$, $p < .001$, $\eta^2_p = 0.814$. Post hoc pairwise comparisons revealed that in the L1 networks, L decreased from 30–40 to 41–50 ($t(198) = 23.53$, $p < .001$, $d = 3.33$), and further to 51–60 ($t(198) = 34.78$, $p < .001$, $d = 4.92$), before rising from 51–60 to 61–70 ($t(198) = -22.31$, $p < .001$, $d = 3.16$). The 51–60 group is characterized by the lowest L among the four groups. The L2 networks showed that L dropped to its lowest in the 41–50 group ($t(198) = 198.91$, $p < .001$, $d = 28.13$), then increased to 51–60 ($t(198) = -86.16$, $p < .001$, $d = 12.19$), with a small decline from 51–60 to 61–70 ($t(198) = 27.89$, $p < .001$, $d = 3.94$). Notably, the largest effect size occurred in the comparison between 30–40 and 41–50 ($d = 28.13$), showing dramatic changes during early-to-mid adulthood. Between-language comparisons revealed that L2 networks had significantly lower L in three of four age groups: 30–40 ($t(198) = 6.42$, $p < .001$, $d = 0.91$), 41–50 ($t(198) = 79.84$, $p < .001$, $d = 11.29$), and 61–70 ($t(198) = 37.56$, $p < .001$, $d = 5.31$). In the 51–60 group, L1 networks exhibited shorter L ($t(198) = -6.67$, $p < .001$, $d = 0.94$), although the effect size was small.

Concerning clustering coefficient (C), L1 networks have consistently higher C than their L2 counterparts across all age groups. The two-way ANOVA revealed significant main effects of age, $F(3, 792) = 1993.222$, $p < .001$, $\eta^2_p = 0.883$, language, $F(1, 792) = 28{,}569.683$, $p < .001$, $\eta^2_p = 0.973$, and an interaction, $F(3, 792) = 620.953$, $p < .001$, $\eta^2_p = 0.702$.

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Post hoc comparisons showed that L1 networks had the lowest C in the 41–50 group. It decreased from 30–40 to 41–50 ($t(198) = 21.80$, $p < .001$, $d = 3.08$), before increasing to 51–60 ($t(198) = -14.71$, $p < .001$, $d = 2.08$), and further to 61–70 ($t(198) = -24.86$, $p < .001$, $d = 3.52$). The L2 networks followed an overall trend similar to L1, but with an additional late-life decline. Specifically, C decreased from 30–40 to 41–50, reaching its minimum ($t(198) = 160.25$, $p < .001$, $d = 22.66$). It then rose to 51–60 ($t(198) = -84.08$, $p < .001$, $d = 11.89$) and declined again to 61–70 ($t(198) = 24.73$, $p < .001$, $d = 3.50$). Regarding between-language differences, L1 networks showed significantly higher C than L2 networks: 30–40 ($t(198) = 55.66$, $p < .001$, $d = 7.87$), 41–50 ($t(198) = 135.91$, $p < .001$, $d = 19.22$, 51–60 ($t(198) = 50.33$, $p < .001$, $d = 7.12$), and 61–70 ($t(198) = 141.77$, $p < .001$, $d = 20.05$).

Finally, regarding modularity (Q), L1 networks have consistently higher Q than L2 across all age groups. The analyses showed significant main effects of age, $F(3, 792) = 1698.889$, $p < .001$, $\eta^2_p = 0.866$, language, $F(1, 792) = 15{,}447.382$, $p < .001$, $\eta^2_p = 0.951$, and an interaction, $F(3, 792) = 1199.759$, $p < .001$, $\eta^2_p = 0.820$. Post hoc comparisons revealed developmental trajectories similar to those observed for clustering coefficient. In the L1 networks, Q decreased significantly from 30–40 to 41–50 ($t(198) = 16.69$, $p < .001$, $d = 2.36$), then increased to 51–60 ($t(198) = -3.73$, $p = .003$, $d = 0.53$), and further to 61–70 ($t(198) = -30.87$, $p < .001$, $d = 4.37$). Notably, the largest increase in adjacent comparisons occurred between 51–60 and 61–70 ($d = 4.37$), representing the most pronounced late-life consolidation of modular structure in L1 lexicon. In the L2 networks, Q declined to its minimum at 41–50 ($t(198) = 117.04$, $p < .001$, $d = 16.55$), then rebounded to 51–60 ($t(198) = -76.86$, $p < .001$, $d = 10.87$). From 51–60 to 61–70, there was a decline ($t(198) = 27.47$, $p < .001$, $d = 3.88$).

Fig. 1 illustrates the developmental trajectories of four network metrics across four age groups in Chinese (L1) and English (L2), including average degree, average shortest path length, clustering coefficient, and modularity. Figs. 2–9 present network visualization for Chinese (L1) and English (L2) among Chinese–English bilingual adults across four age groups (30–40, 41–50, 51–60, 61–70). Each figure illustrates the network for a specific language–age combination. They were constructed in Gephi using the Fruchterman–Reingold layout. Node size and label font size were scaled by degree, and edges were displayed in light gray to improve clarity. The actual response words are shown as node labels, providing a more meaningful interpretation of core and peripheral concepts.

4. Discussion

The findings regarding the age-related development of Chinese word association networks in L1 reveal nonlinear trajectories across adulthood for all key network metrics. Average shortest path length, clustering coefficient, and modularity each followed a U-shaped pattern, reaching a minimum in middle adulthood (primarily in the 41–50 age group, with clustering coefficient and modularity showing lowest values there, and average shortest path length continuing to decline until 51–60 before rebounding in late life). In contrast, average degree

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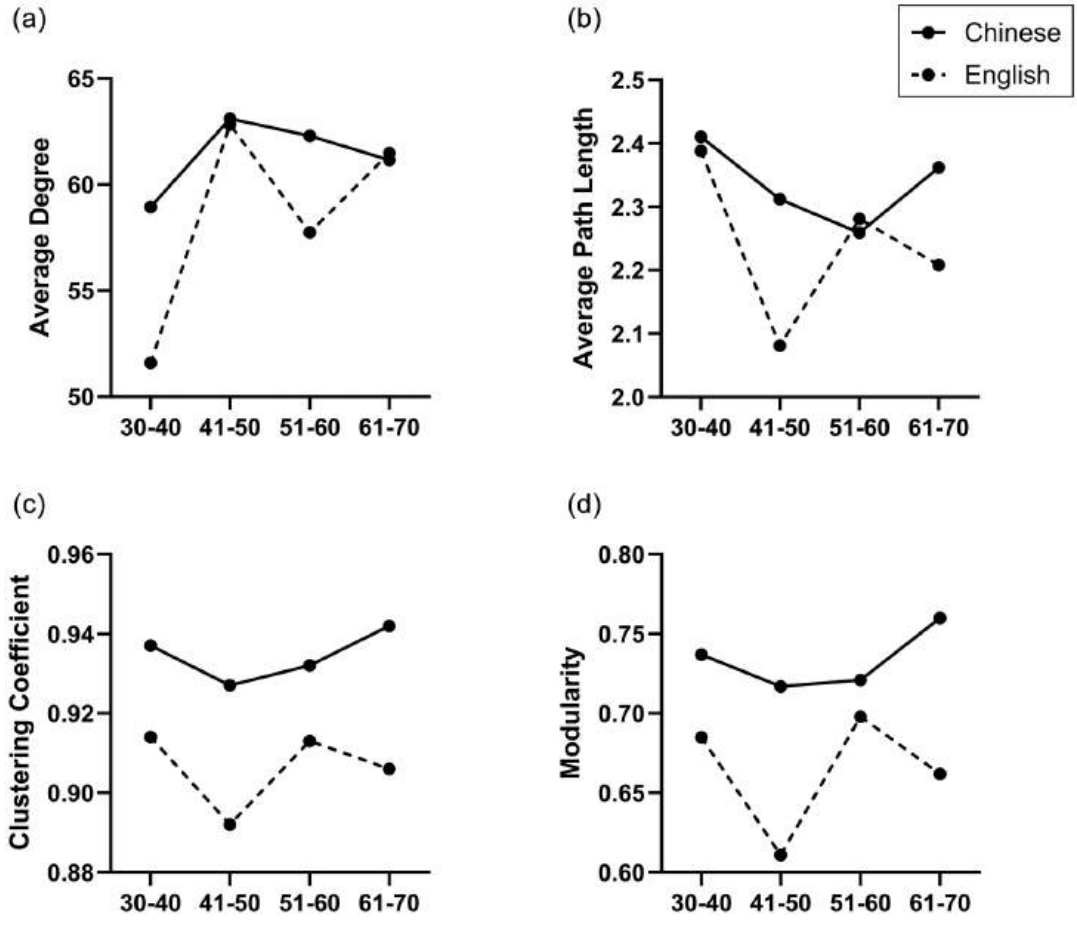


Fig. 1. Developmental trajectories of four network metrics across four age groups in Chinese (L1) and English (L2): (a) average degree; (b) average shortest path length; (c) clustering coefficient; (d) modularity.

exhibited an inverted U-shaped trajectory, with the lowest value in the 30–40 group, peaking in the 41–50 group, and declining thereafter. These dynamics highlight prominent changes during early-to-mid adulthood, particularly between the 30–40 and 41–50 groups, where between-group effect sizes were substantially larger than those observed in other adjacent comparisons. Notably, when analyses are restricted to contrasting early adulthood (30–40) with late adulthood (61–70)—as in some prior studies—older adults exhibit higher average degree, shorter average shortest path length, higher clustering coefficient, and greater modularity, potentially masking the intermediate fluctuations in middle adulthood.

Overall, these observed patterns are partially consistent with prior work. Zortea et al. (2014) compared R–R network structures across children, adults, and older adults, reporting a higher clustering coefficient in older adults than middle-aged adults—similar to our results, suggesting that lexical reorganization in later life may reflect adaptive strategies to main-

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Fig. 2. Chinese network for 30–40 group ($n = 1313$, $m = 38{,}913$).

tain semantic cohesion despite cognitive aging. However, their older group showed longer paths than adults, which contrasts with our findings. This can be attributed to differences in participants' age distributions. While their study compared young adults (mean age = 21.8 years) with significantly older adults (mean age = 70.89 years), our sample spanned middle-aged to older bilinguals (30–70 years) without including younger adults. Crucially, our data revealed an inflection point around age 50, after which average shortest path length began increasing. This trajectory suggests that if we included participants beyond 70 years—as Zortea et al. (2014) did—we might similarly observe average shortest path length exceeding those of younger groups. Additionally, procedural differences may further contribute to

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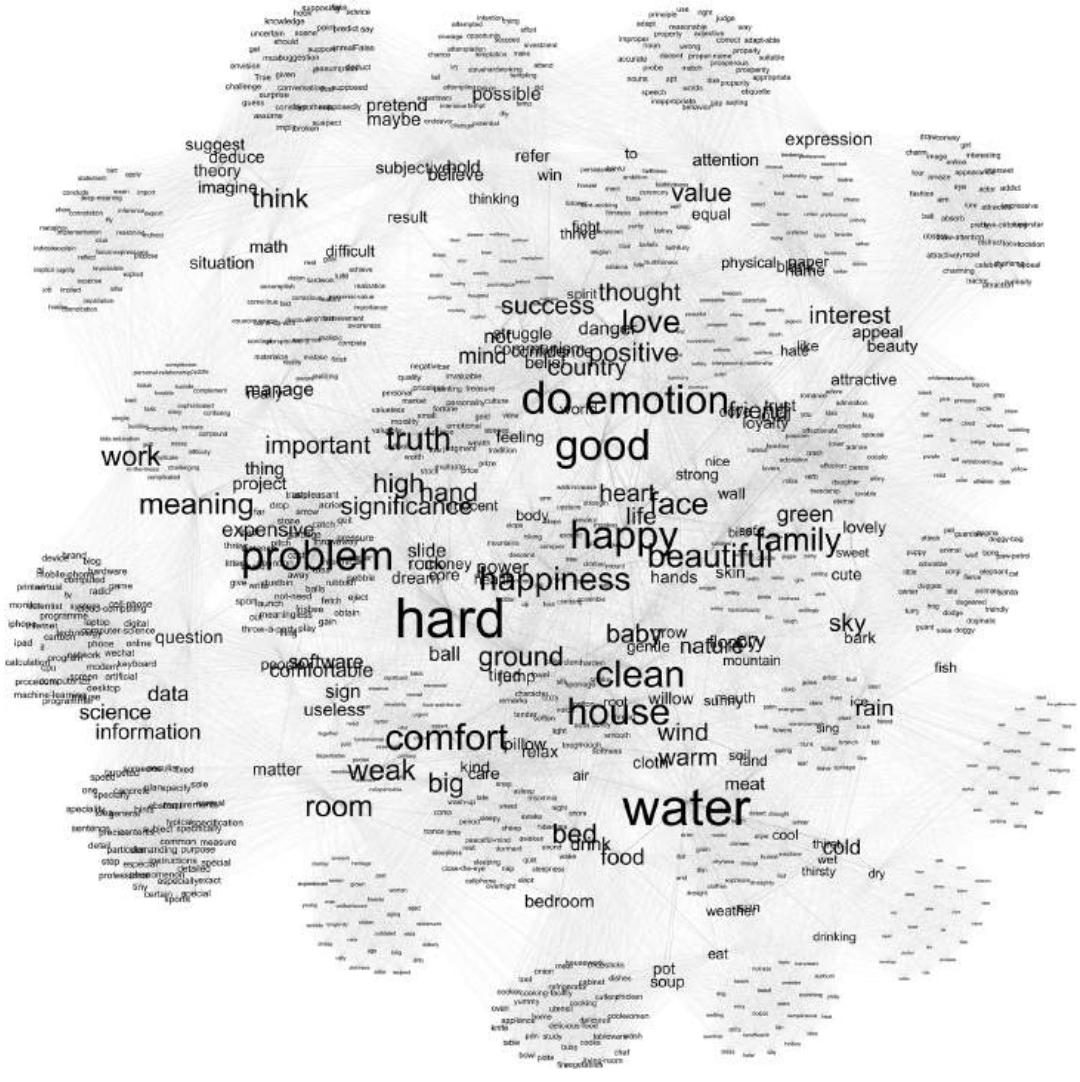


Fig. 3. English network for 30–40 group ($n = 1044$, $m = 27{,}068$).

the discrepancies. In our study, each age-specific subgraph was constrained to 1000 nodes for comparability, derived from 30 cues eliciting three associations each. In contrast, Zortea et al. (2014) used 87 cues with one response per cue, yielding networks with fewer than 700 nodes. This disparity in network scale and response elicitation (multiple vs. single associations) could influence global metrics, particularly average shortest path length, which is sensitive to node count and edge density. Moreover, Zortea et al. (2014) did not employ subgraph bootstrapping or report statistical testing of network metric differences, leaving the significance of young versus older adult contrasts uncertain.

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Fig. 4. Chinese network for 41–50 group ($n = 1350$, $m = 42{,}944$).

In another large-scale study, Dubossarsky et al. (2017) reported patterns that align closely with several of our findings, though with shifts in developmental timing. We observed that average degree increased from ages 30–50, peaking in the 41–50 group before steadily declining—a trajectory broadly consistent with their report of in-degree and out-degree rising through early adulthood, peaking around age 30, and declining thereafter. As for average shortest path length, Dubossarsky et al. (2017) documented a steady increase beginning as

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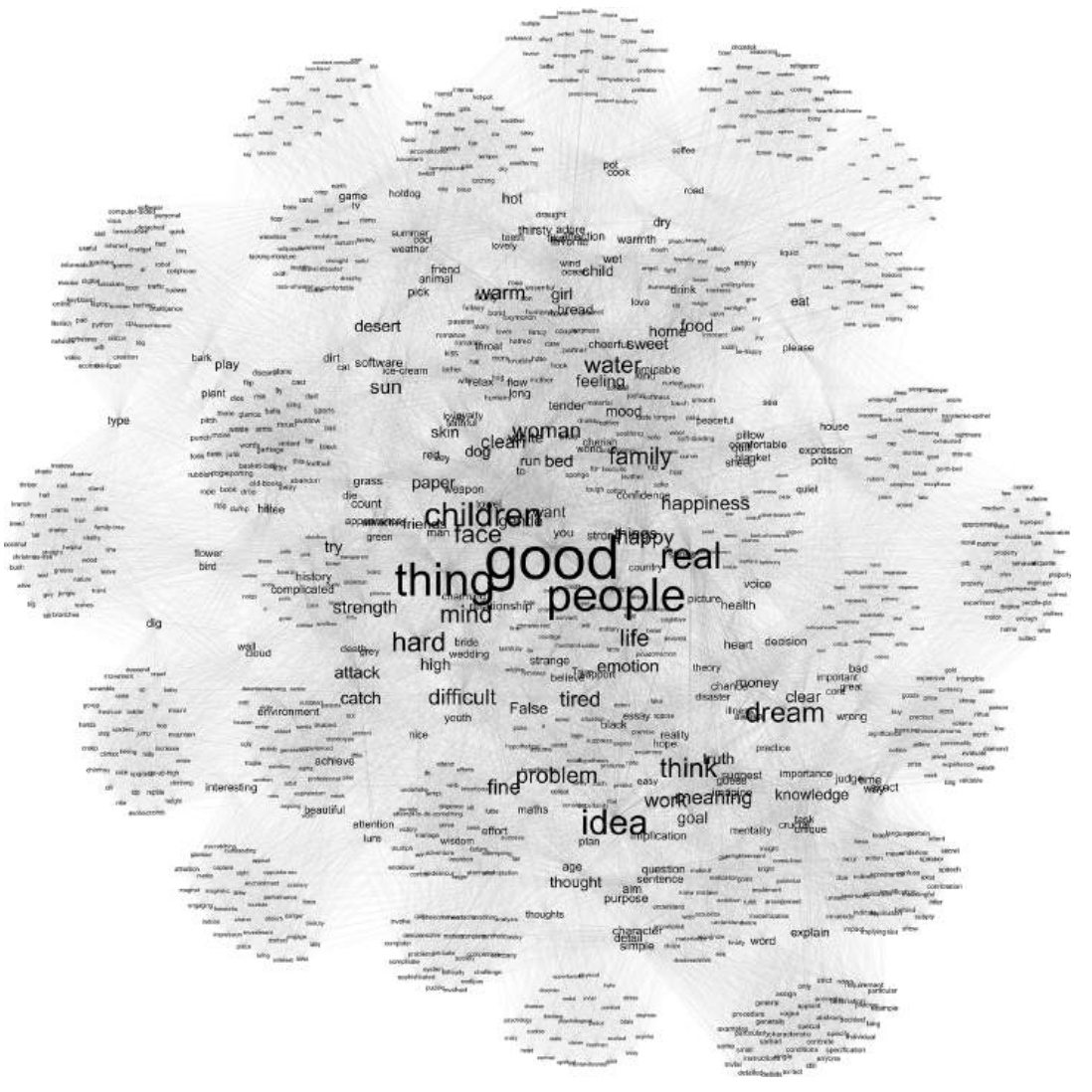


Fig. 5. English network for 41–50 group ($n = 1110$, $m = 35{,}081$).

early as age 30, different from our findings, where the rise was delayed until after age 60. Regarding clustering coefficient, both studies observed a general decline across adulthood, but the timing of the rebound differed: Dubossarsky et al. (2017) reported a modest uptick only in the 60–70 age range, whereas our data showed the increase beginning earlier, from the 41–50 group onward. These differences in timing—particularly the delayed peak in average degree, the postponed onset of increasing path length, and the early onset of increasing clustering coefficient—are likely attributable to sample characteristics. Our participants were exclusively university-level English teachers, a highly educated and cognitively engaged group with sustained, intensive language use. This profile contrasts with the more heteroge-

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Fig. 6. Chinese network for 51–60 group ($n = 1349$, $m = 42{,}135$).

neous population in Dubossarsky et al. (2017), and may have extended the period of network expansion and preserved semantic efficiency into later adulthood. The shared late-life rebound in clustering across studies further supports the idea that compensatory reorganization occurs in older adults who remain linguistically active, regardless of earlier developmental pace.

Jeong and Hills (2024) further demonstrated cross-linguistic variability in aging trajectories by comparing English, Spanish, and Dutch word association networks. For average degree, they reported a monotonic decline across the lifespan in all three languages, which contrasts with our inverted U-shaped trajectory in adulthood (rising to a peak at 41–50 before declining) and with Dubossarsky et al. (2017) lifespan inverted U-shaped pattern. This discrepancy likely arises from procedural differences in network construction: our response−response

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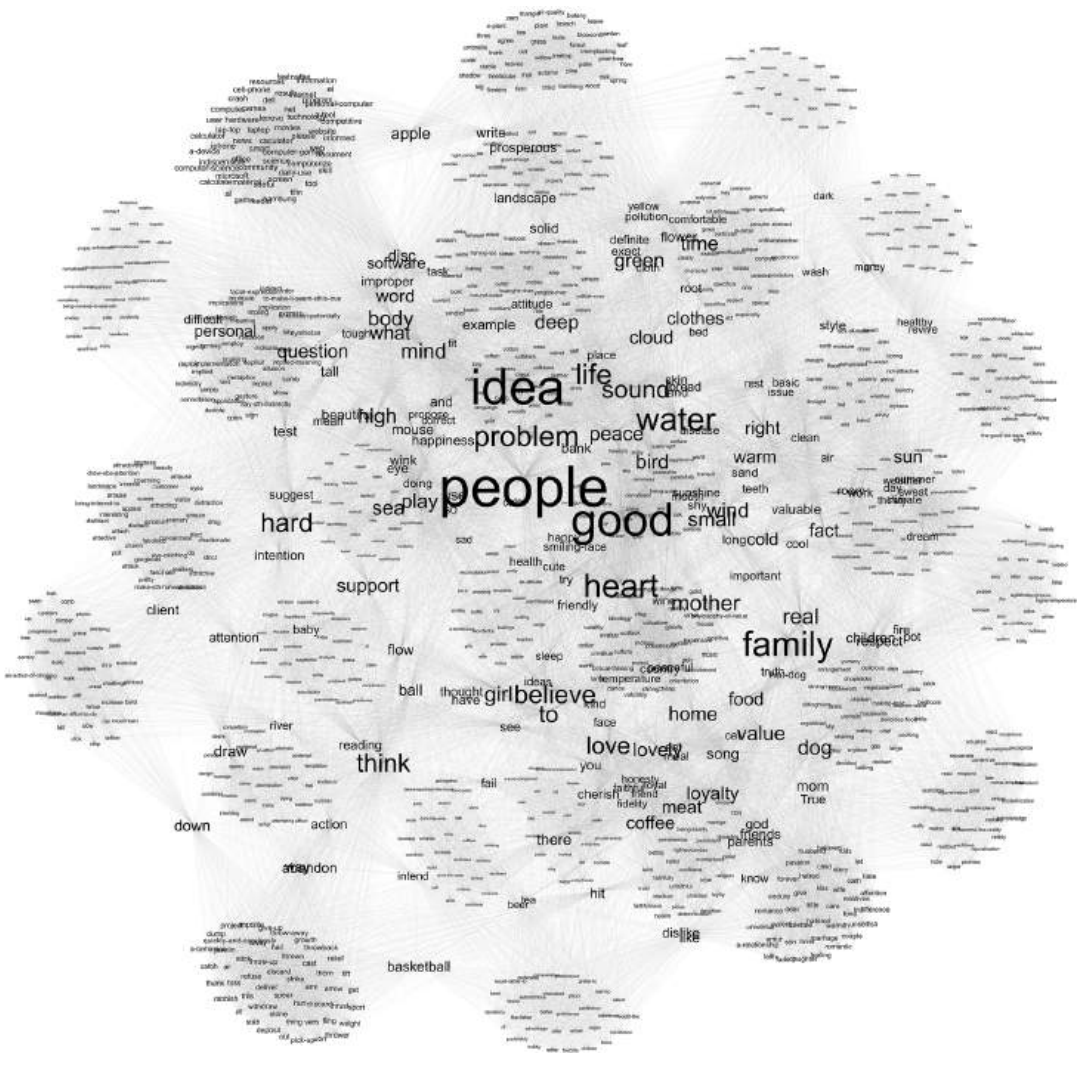


Fig. 7. English network for 51–60 group ($n = 1196$, $m = 35{,}153$).

networks reflect word co-occurrence, Dubossarsky et al. (2017) used cue−cue networks that capture semantic overlap, and Jeong and Hills (2024) combined direct association edges with co-occurrence links. The inclusion of both edge types may reduce sensitivity to changes in associative strength, leading to consistently declining degree values across age. While their regression models suggested a universal increase in average shortest path length and decline in clustering coefficient with age, raw averages revealed language-specific patterns: clustering coefficient in English networks declined slowly before rebounding around age 60—resembling our findings—whereas average shortest path length increased sharply after 60. Our observation of delayed average shortest path length elevation (emerging only after 60, vs. steady decline until then) further reflects the unique profile of our participants,

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Fig. 8. Chinese network for 61–70 group ($n = 1418$, $m = 43{,}919$).

whose sustained educational engagement could buffer typical aging effects on efficiency of semantic organization in the L1 mental lexicon. However, conclusions regarding the structure of semantic networks in late adulthood should be drawn with caution, as within-group variability tends to be greater in older adults than in younger populations. For example, Cosgrove et al. (2021) observed contrasting patterns of semantic network structure between two groups of subjects of similar age range and recruited from the same community, while Wulff et al. (2022) research on individual semantic networks also highlights the pronounced variability within older populations.

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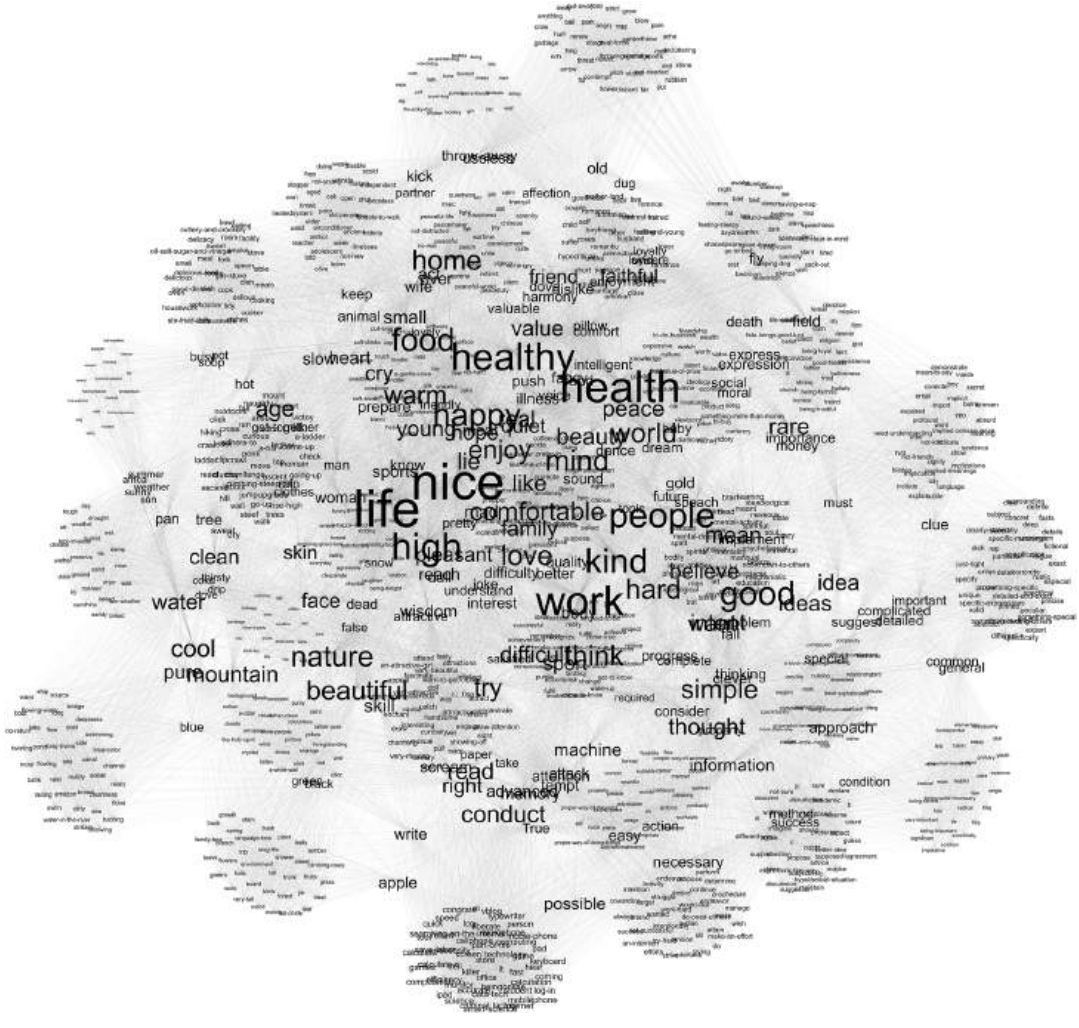


Fig. 9. English network for 61–70 group ($n = 1212$, $m = 37{,}784$).

Furthermore, our findings suggest that network modularity rises with greater magnitude in later adulthood, with the 61–70 group demonstrating substantially higher modularity than other age groups. This observation mirrors findings from Cosgrove et al. (2021) and Cosgrove et al. (2023), in which older adults (typically aged 60+) showed notably higher modularity than younger groups. The late-life increase in modularity renders support for theoretical accounts proposing compensatory mechanisms in aging, such as the compensation-related utilization of neural circuits hypothesis (Reuter-Lorenz & Cappell, 2008), which posits that older adults develop more segregated neural subsystems to maintain cognitive function. The trajectory observed here—relatively stable modularity through midlife followed by post-60 increases—suggests this compensatory mechanism may become particularly salient after retirement age. However, this pattern could also result from older adults producing more

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responses within the same semantic group, due to reduced ability to switch categories. For example, when given the cue *shuì* (睡, "sleep"), the 61–70 age group often generated tightly related bedroom items—*chuáng* (床, "bed"), *bèizi* (被子, "quilt"), *wēndù* (温度, "temperature")—staying within a single "bedroom/sleep environment" cluster. In contrast, younger groups switched more freely across clusters: the 51–60 group produced *chuáng* (床, "bed"), *mèng* (梦, "dream"), *hēiyè* (黑夜, "night"); the 41–50 group gave *hāqian* (哈欠, "yawn"), *zhěntou* (枕头, "pillow"), *chuāng* (窗, "window"); and the 30–40 group offered *kùn* (困, "sleepy"), *mènghuà* (梦话, "sleep-talking"), *shūfu* (舒服, "comfortable"). This age-related tendency to remain in one cluster densifies intramodule connections and increases modularity even if the underlying lexical network stays the same. This modularity pattern holds implications for creativity, as recent research (Campidelli et al., 2026) has shown that individuals with lower modularity and shorter path lengths exhibit higher creative performance. In our study, these metrics reach their minimum during middle adulthood—average shortest path length at its lowest in the 51–60 group and modularity in the 41–50 group—suggesting that creativity may peak among bilingual adults in midlife.

Another important finding of this study concerns the systematic structural differences between L1 and L2 semantic networks. Overall, the L1 networks exhibited higher average degrees, longer paths, higher clustering, and greater modularity than L2 networks. These patterns both align with and diverge from previous bilingual network studies. For example, Borodkin et al. (2016) examined English–Hebrew bilingual semantic network using a semantic fluency task and found that L1 networks tend to have longer path lengths and higher modularity. Thus, it can be assumed that L1 semantic systems are more community-structured, which may provide advantages for flexible and efficient lexical retrieval in spontaneous language use. However, contrary to Borodkin et al. (2016) observation, our data revealed significantly higher clustering coefficients in L1 networks. This discrepancy may originate from differences in participant demographics and network construction methods. Their participants were undergraduates, and the cues were drawn from restricted semantic categories (e.g., *fruits*, *vegetables*, and *animals*) with networks capped at fewer than 100 nodes. In comparison, our participants were all over the age of 30 and working as university English teachers. More critically, the networks were derived from a free word association task and comprised of over 1000 nodes, which allowed for a more spontaneous and unconstrained representation of semantic networks. Our findings on path length and clustering are consistent with more recent work by Agustín-Llach and Palapanidi (2024) on Greek–Spanish bilinguals, and Martinez and Vitevitch (2024), who compared L1 English and L2 Kaqchikel semantic networks. These studies, along with ours, suggest that the enhanced global and local connectivity observed in L1 semantic networks is robust across both younger and older adult populations.

Importantly, the findings reveal that, across both L1 and L2, and for all network metrics, middle adulthood is marked by dynamic and even unpredictable fluctuations, with these changes being particularly pronounced in L2. For instance, average degree surges from the 30–40 to 41–50 group before sharply declining, average shortest path length first decreases substantially, then rises, while clustering coefficient and modularity both drop markedly before increasing. These substantial shifts in average degree and average shortest path length may reflect the expansion and improved connectivity of the L2 lexicon during the period of

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career establishment and advancement, followed by subsequent reorganization in mid-to-late adulthood. We posit that such fluctuations arise from multiple factors, chief among them being language learning and use influenced by social contexts and professional needs, alongside cognitive decline caused by aging. This perspective aligns with contemporary views in cognitive aging and bilingualism research (e.g., Hills, 2025b; Ramscar, Sun, Hendrix, & Baayen, 2017; Reifegerste, 2021), which frame aging not as mere decline but as a product of ongoing learning and enrichment. Furthermore, this underscores that the bilingual aging lexicon is not a simplistic young-versus-old dichotomy but a continuous process, where middle age often yields highly informative insights (Reifegerste, 2021) that warrant greater attention.

Regarding generalizability, our findings are most directly applicable to highly educated, aging bilingual populations—a group of growing interest in the field of bilingualism and cognitive aging. The observed stable L1–L2 differences across the adult lifespan align closely with behavioral and neurocognitive evidence from related domains. For instance, Reifegerste, Elin, and Clahsen (2019) reported persistent processing advantages for L1 over L2 in inflectional morphology among older adults, mirroring our L1 > L2 pattern in network metrics. Similarly, Reifegerste et al. (2020) demonstrated that early educational attainment preserves declarative memory in late life—a factor relevant to our highly educated cohort. At the neural level, DeLuca and Voits (2022) found that bilingual experience attenuates age-related decline in white matter integrity, a key substrate for efficient neural communication. This may underlie our observation that average shortest path length continued to decrease through midlife (rather than rising monotonically as in monolingual studies), with clustering coefficients increasing into late life—patterns suggestive of compensatory or protective network reorganization. Recent intervention evidence further supports the view that language learning and use can serve as a pathway to cognitive maintenance in later life. For example, Wen and Dong (2025) showed that even short-term, between-dialect interpreting training led to measurable gains in working memory, interference control, and cognitive flexibility among older adults. Such findings reinforce the broader implication of our results—that sustained bilingual engagement may contribute to preserved or even enhanced semantic organization across aging.

However, several limitations must be noted. First, our use of response–response networks, while effective for capturing group-level patterns, limits direct comparability with results from cue–response or cue–cue networks due to differences in network construction. Future research should construct and compare these network types using the same dataset to evaluate methodological impacts. Future research should, therefore, construct and compare multiple network types using the same dataset to better evaluate the methodological impact of different network representations. In addition, although the present study employed a BFS subgraph sampling approach to generate size-controlled subnetworks for statistical comparison, alternative robustness checks—such as cue-word resampling or other graph-sampling strategies—could further assess the generalizability of observed network patterns across different stimulus subsets.

Second, our study targeted a well-controlled cohort of Chinese–English bilingual educators and elicited word associations in both languages, a design that necessarily differs from large-scale normative projects involving thousands of heterogeneous participants (e.g., De

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Deyne, Navarro, Perfors, Brysbaert, & Storms, 2019). While this approach allowed us to examine bilingual lexical organization within a tightly matched population, the resulting sample size—especially in the older groups—is smaller than that of large norms. Future research could extend this work by combining similarly well-characterized bilingual samples with larger-scale data collection efforts, enabling complementary insights that leverage both methodological control and broader population coverage.

Additionally, because our analyses were conducted at the aggregate level, we were not able to examine within-individual relationships between L1 and L2 network structures. Recent research highlights that findings from group-based semantic network analyses can diverge from those obtained at the individual level, particularly in aging populations (Cosgrove, Beaty, Sandberg, Kenett, & Diaz, 2026). This underscores the importance for future studies to integrate both approaches in their designs and interpretations—not to replace the group network approach, which remains invaluable for identifying overarching developmental trajectories, but to complement them with individualized methods for a fuller understanding of bilingual lexical dynamics. Future work using such hybrid individualized network approaches may help clarify how L1 and L2 lexical structures relate within individuals over time.

Finally, it should be noted that the present study relied on participants' self-reports to confirm the absence of cognitive impairment. While this approach is consistent with previous large-scale lexical aging research, it may not fully capture mild or undiagnosed cognitive decline, particularly among participants over 60 years old. Future studies should, therefore, include standardized cognitive assessments to ensure a more reliable evaluation of cognitive health in aging populations.

5. Conclusion

In conclusion, this study employed a network approach to examine age-related and cross-linguistic differences in the mental lexicon of Chinese–English bilingual adults. By analyzing word association networks across four age groups, the results reveal distinct structural patterns in L1 (Chinese) and L2 (English) lexical networks across adulthood. Specifically, the findings show systematic differences between the two languages in global connectivity, clustering, and modular organization, as well as age-related changes in these network properties over time. These results provide new descriptive evidence on how lexical network structure varies across the adult lifespan within bilingual speakers. More broadly, the study demonstrates the value of network-based approaches for investigating lexical organization and contributes to ongoing efforts to characterize structural properties of bilingual lexical systems using behavioral association data.

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Conflicts of interest

The authors have no conflicts of interest to declare.

Data availability statement

The data that support the findings of this study are openly available in the Open Science Framework at https://osf.io/t6wsb/.

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Supporting Information

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Supplementary Information

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